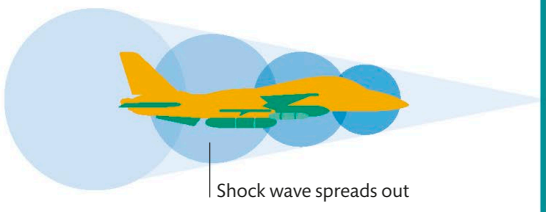
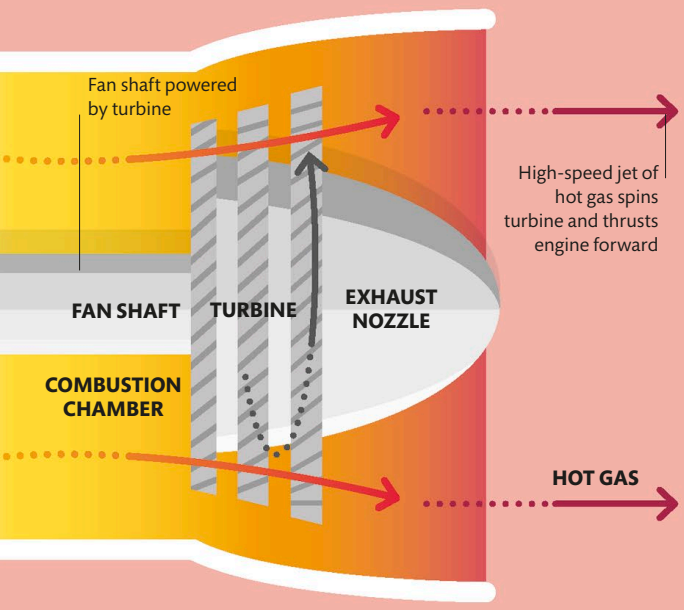
THE SOUND BARRIER

Planes flying faster than the speed of sound compress the air in front of them so much that they form a high-pressure shock wave. This spreads out and is heard on the ground as a loud sonic boom.





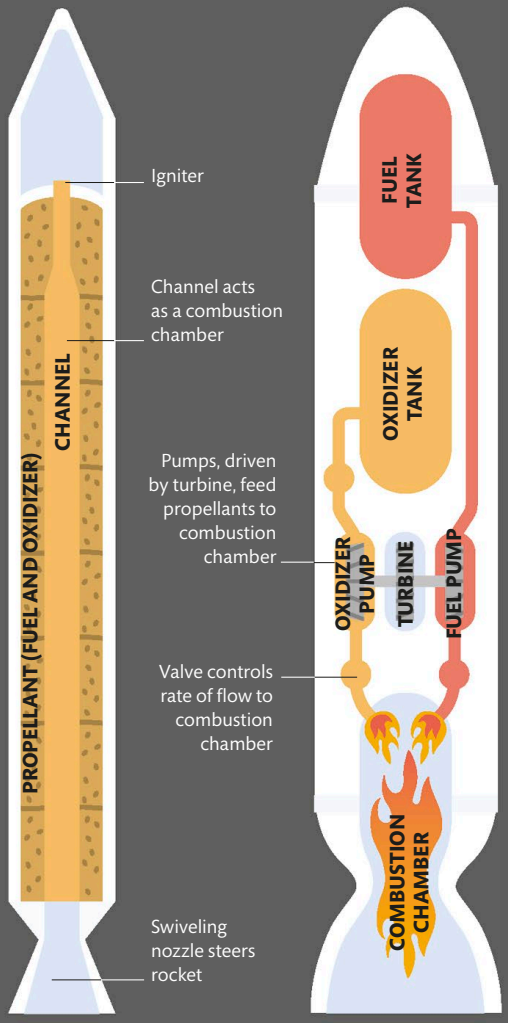
4 Turbine
The hot gas expands explosively and rushes out of the engine, spinning the blades of a turbine. The spinning turbine powers the fan and compressor.

5 Exhaust nozzle
The jet of hot exhaust gas leaves the engine, along with the cold bypass air, pushing back against the engine, and generating thrust.

THE SUPERSONIC AIRLINER **CONCORDE** FLEW FROM **NEW YORK TO LONDON** IN **2 HOURS AND 52 MINUTES**

Rocket engines

Unlike jet engines, which use oxygen from the atmosphere to burn their fuel, rockets carry their own oxygen supply, which means that they can operate in the vacuum of space. The oxygen supply, or oxidizer, can take the form of pure liquid oxygen or an oxygen-rich chemical compound.



Solid-fueled rocket
The fuel and oxidizer are mixed together as a solid compound with an open channel in the middle. When the igniter fires, the fuel burns along the channel until there is none left.

Liquid-fueled rocket
The fuel and oxidizer are stored as liquids. Unlike a solid-fueled rocket, a liquid-fueled rocket can be restarted. It can also be throttled by varying the flows of fuel and oxidizer.